Taming Infinity

A Step-by-Step Walkthrough of Renormalization
Applied to Vacuum Polarization in QED

Particle Physics Reading Guide — Griffiths §7.9

1 What We Start With

We have computed the amplitude for electron–muon scattering including a vacuum polarization loop. The result, from the Feynman rules, is (Griffiths Eq. 7.176):

$$\mathcal{M} = \frac{-ig_e^4}{q^4}\left[\bar{u}(p_3)\gamma^\mu u(p_1)\right]\left\{\int \frac{d^4k}{(2\pi)^4}\,\frac{\mathrm{Tr}\left[\gamma_\mu(\not{k}+mc)\,\gamma_\nu(\not{k}-\not{q}+mc)\right]}{(k^2-m^2c^2)\left[(k-q)^2-m^2c^2\right]}\right\}\left[\bar{u}(p_4)\gamma^\nu u(p_2)\right] \tag{1}$$

The external fermion currents are perfectly well-behaved. All the trouble is concentrated in the integral in curly braces. Our job is to make sense of it.

The Mission

We have a divergent integral sitting inside an otherwise sensible amplitude. We need to extract a finite, physically meaningful prediction from it. That's what renormalization does.

The loop integral, stripped of the external pieces, defines the object:

$$\boxed{I_{\mu\nu} = -g_e^2\int \frac{d^4k}{(2\pi)^4}\,\frac{\mathrm{Tr}\left[\gamma_\mu(\not{k}+mc)\,\gamma_\nu(\not{k}-\not{q}+mc)\right]}{(k^2-m^2c^2)\left[(k-q)^2-m^2c^2\right]}} \tag{2}$$

This is our raw ingredient. Everything that follows is the procedure for cooking it.

2 Step 1: Determine What Shape the Answer Can Have

Step 1 — Tensor Decomposition

Before doing any integral, figure out what the answer is *allowed* to look like based on Lorentz symmetry alone.

$I_{\mu\nu}$ carries two Lorentz indices — it's a rank-2 tensor. Once we integrate over k, the loop momentum k disappears entirely (it's a dummy integration variable, like x after computing $\int f(x)\,dx$). The only four-vector that survives is q^μ, the external photon momentum.

So we ask: what rank-2 tensors can you build from a single four-vector q^μ and the metric $g_{\mu\nu}$? There are exactly two:

$$g_{\mu\nu} \qquad \text{and} \qquad q_\mu q_\nu \tag{3}$$

That's it. There is nothing else available. So regardless of how complicated the integral is, the answer *must* take the form:

$$I_{\mu\nu} = A(q^2)\, g_{\mu\nu} + B(q^2)\, q_\mu q_\nu \tag{4}$$

where A and B are scalar functions that can only depend on q^2 (the only Lorentz scalar you can build from q^μ).

Why This Step Matters

This is not a calculation — it's a constraint from symmetry. We've reduced an intimidating tensor integral to the problem of finding two scalar functions. Think of it like this: if someone hands you a box and says "the contents weigh 10 pounds," you don't need to open the box to know the weight. Lorentz symmetry is telling us the shape of the box before we open it.

Griffiths writes this with conventional prefactors as (Eq. 7.180):

$$I_{\mu\nu} = -ig_{\mu\nu}\, q^2\, I(q^2) + q_\mu q_\nu\, J(q^2) \tag{5}$$

The $-i$ and extra q^2 in the first term are just pulled out for later algebraic convenience. The content is identical.

3 Step 2: Kill What Doesn't Contribute

Step 2 — Eliminate Irrelevant Pieces

Use physical constraints to throw away tensor structures that can't affect the final answer.

Look at how $I_{\mu\nu}$ appears in the full amplitude. The index μ contracts with γ^μ in the electron current $\bar{u}(p_3)\gamma^\mu u(p_1)$, and ν contracts with γ^ν in the muon current $\bar{u}(p_4)\gamma^\nu u(p_2)$.

The $q_\mu q_\nu$ term produces factors like:

$$q_\mu\, \bar{u}(p_3)\gamma^\mu u(p_1) = \bar{u}(p_3)\, \not{q}\, u(p_1) \tag{6}$$

Now, $q = p_1 - p_3$ (momentum conservation at the electron vertex). Using the Dirac equation:

$$\bar{u}(p_3)\, \not{q}\, u(p_1) = \bar{u}(p_3)(\not{p}_1 - \not{p}_3)u(p_1) = \bar{u}(p_3)(mc - mc)u(p_1) = 0 \tag{7}$$

So the entire $q_\mu q_\nu$ piece of $I_{\mu\nu}$ vanishes when plugged into the physical amplitude. We can forget $J(q^2)$ completely.

The Payoff

We started with a tensor integral that could in principle yield two independent scalar functions. Symmetry told us the general form (Step 1), and the equations of motion killed one of the two pieces (Step 2). We're now down to a single unknown function: $I(q^2)$.

4 Step 3: Evaluate the Surviving Function

Step 3 — Do the Integral (and Face the Divergence)

Evaluate $I(q^2)$. This is the hard, technical part. The result will contain an infinity.

This step involves serious integral calculus — Feynman parametrization, Wick rotation, and careful bookkeeping. Griffiths quotes the result (combining Eqs. 7.182–7.186):

$$I(q^2) = \frac{1}{12\pi^2}\left[\ln\left(\frac{M^2}{m^2c^2}\right) - f\left(\frac{-q^2}{m^2c^2}\right)\right] \tag{8}$$

Here M is a momentum cutoff: we temporarily capped the integral at $|k| = M$ to keep it finite. The function f is given explicitly by Griffiths (Eq. 7.184) and is perfectly finite for all values of its argument.

Let's stare at this result. It has two pieces:

Piece	**Expression**	**Character**
First	$\frac{1}{12\pi^2}\ln\left(\frac{M^2}{m^2c^2}\right)$	Diverges as $M \to \infty$. **No q^2 dependence.**
Second	$\frac{1}{12\pi^2} f\left(\frac{-q^2}{m^2c^2}\right)$	Perfectly finite. **Depends on q^2.**

Why the Divergence Is q^2-Independent

The divergence comes from the region of integration where the loop momentum k is enormous — much larger than the external momentum q. When $|k| \gg |q|$, the propagator denominators become:

$$k^2 - m^2c^2 \quad \text{and} \quad (k-q)^2 - m^2c^2 \;\approx\; k^2 - m^2c^2$$

The integrand can't "see" q anymore. A tidal wave doesn't notice a pebble. So the contribution from this region — which is where the integral blows up — is the same number regardless of what q^2 is. The divergence is a universal constant.

This single fact is what makes renormalization possible. If the divergence depended on q^2, you couldn't separate it from the physics, and the whole procedure would fail.

5 Step 4: See What the Loop Did to the Propagator

Step 4 — Interpret the Result Physically

Before doing any renormalization, understand what the loop correction means for the photon propagator.

The tree-level amplitude has the photon propagator $-ig_{\mu\nu}/q^2$ sitting between the two fermion currents. Including the loop modifies this to (Griffiths Eq. 7.177):

$$\frac{-ig_{\mu\nu}}{q^2} \;\longrightarrow\; \frac{-ig_{\mu\nu}}{q^2} - \frac{i}{q^4} I_{\mu\nu} = \frac{-ig_{\mu\nu}}{q^2}\Big[1 + I(q^2)\Big] \tag{9}$$

So the vacuum polarization loop *dresses* the photon propagator. The photon no longer travels cleanly between the two charges — it spends part of its time as a virtual e^+e^- pair. This modifies the effective strength of the interaction.

Substituting our expression for $I(q^2)$, the full amplitude becomes (Griffiths Eq. 7.188):

$$\mathcal{M} = \frac{-g_e^2}{q^2}\big[\bar{u}(p_3)\gamma^\mu u(p_1)\big]\left\{1 + \frac{g_e^2}{12\pi^2}\left[\ln\left(\frac{M^2}{m^2c^2}\right) - f\left(\frac{-q^2}{m^2c^2}\right)\right]\right\}\big[\bar{u}(p_4)\gamma_\mu u(p_2)\big] \tag{10}$$

This is a correct expression for the amplitude — but it contains M, which is fictitious. We need to get rid of it. That's where renormalization enters.

6 Step 5: Define the Physical Charge

Step 5 — Absorb the Infinity into a Measured Quantity

Define a "renormalized" coupling constant that absorbs the divergent piece. This is the coupling you actually measure in the lab.

Here is the critical move. We define the **renormalized coupling constant**:

$$\boxed{g_R^2 \equiv g_e^2\left[1+\frac{g_e^2}{12\pi^2}\ln\left(\frac{M^2}{m^2c^2}\right)\right]} \tag{11}$$

What is this, physically? It's the effective coupling that *includes* the effect of vacuum polarization evaluated at $q^2=0$. And $q^2=0$ corresponds to a static, long-distance Coulomb measurement — exactly the kind of experiment Millikan did to measure e. So g_R is the charge you actually find in the lab.

The bare coupling g_e was the parameter we naively wrote into the Lagrangian, pretending there were no quantum corrections. Nobody has ever measured it. It was always an intermediate bookkeeping device.

The Conceptual Leap

We are *not* computing the charge from first principles (that would give infinity). We are instead saying: "I measured the charge in the lab and got g_R. Let me use that measured value as my input, and predict what happens at other energies."
This is exactly what we do in every other area of physics. You don't derive G (Newton's gravitational constant) from theory — you measure it, plug it in, and predict orbits. Renormalization is the same idea, applied to a situation where the naive "derivation" gives infinity.

7 Step 6: Rewrite Everything in Terms of g_R

Step 6 — Eliminate the Cutoff and Obtain a Finite Prediction

Express the amplitude in terms of the physical coupling g_R. The cutoff M will cancel.

From the definition of g_R, we can solve for g_e^2:

$$g_e^2 = g_R^2\left[1-\frac{g_R^2}{12\pi^2}\ln\left(\frac{M^2}{m^2c^2}\right)\right]+\mathcal{O}(g_R^6) \tag{12}$$

(We only need this to the order we're working at — the correction is already of order g_e^4, so replacing $g_e \to g_R$ inside the correction term introduces errors of still higher order, which we consistently neglect.)

Now substitute into the amplitude. Everywhere we had g_e^2, we write it in terms of g_R^2. The $\ln M$ from the loop and the $\ln M$ from the definition of g_R meet each other and cancel:

$$\underbrace{\frac{g_e^2}{12\pi^2}\ln\frac{M^2}{m^2c^2}}_{\text{from the loop}} - \underbrace{\frac{g_R^2}{12\pi^2}\ln\frac{M^2}{m^2c^2}}_{\text{from replacing } g_e \text{ with } g_R} = 0 + \mathcal{O}(g^6) \tag{13}$$

What survives is:

The Renormalized Amplitude (Griffiths Eq. 7.190)

$$\mathcal{M} = \frac{-g_R^2}{q^2}\left[\bar{u}(p_3)\gamma^\mu u(p_1)\right]\left\{1 + \frac{g_R^2}{12\pi^2} f\left(\frac{-q^2}{m^2c^2}\right)\right\}\left[\bar{u}(p_4)\gamma_\mu u(p_2)\right] \tag{14}$$

Look at what we have:

- **No M anywhere.** The fictitious cutoff has completely disappeared. The expression is independent of the regulator.
- **g_R is a measured number.** It's the fine structure constant at $q^2 = 0$, determined by experiment: $\alpha = g_R^2/4\pi \approx 1/137$.
- **$f(-q^2/m^2c^2)$ is a finite, calculable function.** It encodes how the effective interaction strength changes with momentum transfer.
- **The theory is predictive.** Given one measurement (g_R), we can compute the scattering amplitude at any q^2.

8 Step 7: The Running Coupling

Step 7 — Extract the Physical Consequence

The finite leftover from renormalization tells us that the effective charge depends on the momentum scale at which you probe it.

We can absorb the finite correction into the coupling constant itself, promoting it to a q^2-dependent quantity (Griffiths Eq. 7.192):

$$\boxed{\alpha(q^2) = \frac{\alpha(0)}{1 - \dfrac{\alpha(0)}{3\pi} f\left(\dfrac{-q^2}{m^2c^2}\right)}} \tag{15}$$

This is the **running coupling constant**. It tells us:

- At $q^2 = 0$ (long distance, low energy): $\alpha = 1/137$, the familiar textbook value.
- At $|q^2| \gg m^2c^2$ (short distance, high energy): f grows logarithmically, so α increases. You're probing inside the screening cloud of virtual pairs.
- At $q^2 \sim M_Z^2$ (the Z boson mass): $\alpha \approx 1/128$, which has been experimentally measured and confirmed.

The Punchline

The "constant" 1/137 is only the long-distance, low-energy value of a quantity that actually varies with the scale at which you measure it. The variation is tiny in everyday physics (corrections of order 10^{-6}), which is why nobody noticed it for centuries. But at high energies, it's real, measurable, and confirmed by experiment.
Renormalization didn't just save us from an infinity — it revealed genuine new physics: the electromagnetic coupling is not a fixed number, but a function of the resolution at which you examine nature.

9 The Whole Procedure at a Glance

The Renormalization Recipe — Summary

1. **Tensor decomposition.** Use Lorentz symmetry to constrain the form of the loop integral. Reduce it to unknown scalar functions.

2. **Kill irrelevant pieces.** Use equations of motion / current conservation to eliminate tensor structures that don't contribute to the physical amplitude.

3. **Evaluate.** Compute the surviving scalar function. Regulate the divergence (cutoff, dimensional regularization, etc.). Get: (infinite constant) + (finite function of q^2).

4. **Interpret.** Understand the loop as a modification of the propagator. The interaction now has a q^2-dependent correction on top of the bare piece.

5. **Define the physical parameter.** Declare that the measured coupling absorbs the infinite constant. This is not a trick — it's the recognition that the bare parameter was never observable.

6. **Rewrite in terms of the physical parameter.** The infinity cancels. What remains is finite, cutoff-independent, and expressed entirely in terms of measurable quantities.

7. **Extract the physics.** The finite leftover gives the running coupling — a prediction of how the interaction strength changes with energy scale.

Why It Works — The Deepest Reason

The divergence is independent of external kinematics. The finite leftover depends on external kinematics. Because they live in different worlds — one is a universal constant, the other is a function of q^2 — you can cleanly separate them. The constant gets absorbed into a measured parameter. The function becomes a prediction. That separation is not a trick. It's forced on you by the mathematical structure of the integral, which in turn reflects the fact that ultra-short-distance physics (where the divergence lives) is decoupled from the longer-distance physics (where your experiment lives).